

Healthcare Appointment System

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1. A brief project summary (i.e what does your application do?)

The Healthcare Appointment System is a comprehensive application designed to streamline the scheduling and management of medical appointments between patients and healthcare providers. It allows patients to easily book, modify, or cancel appointments with doctors across various specialties. Doctors can efficiently manage their schedules, maintain records of treatments provided, and issue prescriptions. Furthermore, the system tracks treatment histories, enabling both patients and healthcare providers to easily access past treatment information and prescribed medications, ensuring continuity and quality of patient care.

2. A tentative business rules and restrictions that your database might have

1. Patients:

- Each patient must have a unique identifier (patient_id).
- A patient's email address must be unique to prevent duplicate patient records.
- Patients must provide valid contact information (phone or email) upon registration.

2. Doctors:

- Each doctor must have a unique identifier (doctor_id).
- Doctors must be associated with one specific department and medical specialty.
- Doctors' email addresses must be unique.

3. Appointments:

- Each appointment must involve exactly one patient and one doctor.
- Appointment dates and times cannot overlap for the same doctor.
- Appointments cannot be scheduled for past dates and times.
- The status of an appointment can only be 'Scheduled', 'Completed', or 'Canceled'.

4. Treatment History:

- Each treatment record must link to an existing appointment.
- Treatment history records are created only for appointments marked as 'Completed'.
- Treatment records must contain diagnosis and treatment details.
- Optionally indicates whether follow-up is required.

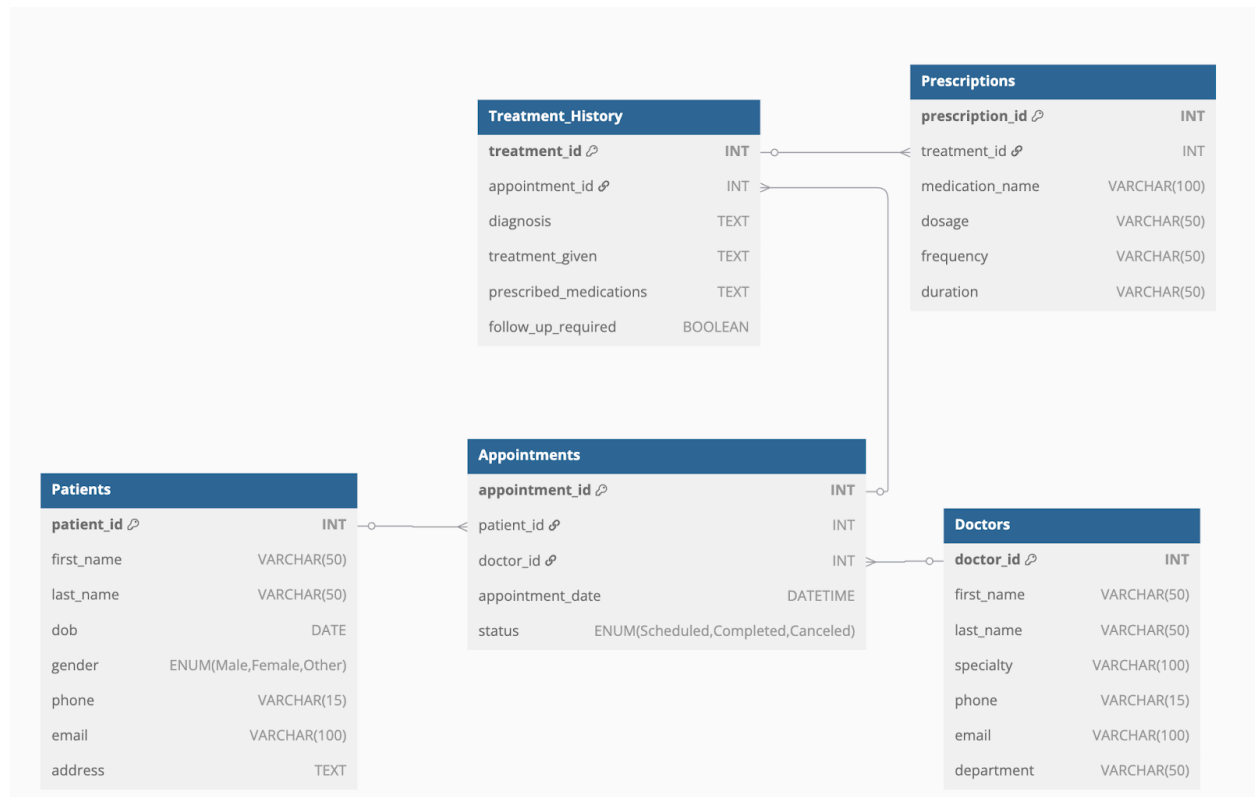
5. Prescriptions:

- Every prescription record must originate from a valid treatment history entry.
- Prescriptions must specify medication details clearly, including dosage, frequency, and duration.

General Constraints:

- Deleting a patient or doctor should be restricted if they have upcoming scheduled appointments.
- Canceling an appointment updates its status to 'Canceled' but does not delete its record.
- Past appointments marked as 'Completed' cannot be deleted or modified except by authorized personnel.
- Data integrity must be maintained by enforcing referential integrity between related tables.

3. A tentative ER diagram showing the conceptual model for the DB



4. A schema representation for the ER diagram

PATIENTS

first_name	last_name	dob	<u>patient_id</u>	gender	phone	email	address
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DOCTORS

first_name	last_name	specialty	<u>doctor_id</u>	phone	email	department
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APPOINTMENTS

appointment_date	doctor_id	<u>appointment_id</u>	patient_id	status
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TREATMENT_HISTORY

diagnosis	treatment_given	<u>treatment_id</u>	appointment_id	prescribed_medications	follow_up_required
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PRESCRIPTION

dosage	medication_name	<u>prescription_id</u>	treatment_id	frequency	duration
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